

A scalar counterexample to property (P') as printed in arXiv:1003.0108

Solution packet for `fa_banach_001`

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Status. Full negative answer to the literal statement of (P') in Section 6. The counterexample already lies in the one-point scalar rational setting.

The printed question

Ball and Sasane define the stability margin

$$\mu_{P,C} = \|H(P, C)\|_{\infty}^{-1}$$

when C stabilizes P , and define it to be zero otherwise. Their final section asks whether the following property always holds in the abstract setup:

(P') : C satisfying $\mu_{P_0,C} > m$ stabilizes P only if $d_{\nu}(P, P_0) \leq m$.

Read literally, this says

$$\mu_{P_0,C} > m \quad \text{and} \quad C \text{ stabilizes } P \quad \implies \quad d_{\nu}(P, P_0) \leq m. \tag{1}$$

Theorem 1. *Implication (1) is false under assumptions (A1)–(A4), even with $R = S = \mathbb{C}$, scalar plants, and $m = \frac{1}{2}$.*

Proof. Take $R = S = \mathbb{C}$ with complex conjugation as involution, take the index group to be trivial, and let $\iota = 0$. This satisfies (A1)–(A4): $\mathbb{C}$ is a unital integral domain and a semisimple commutative Banach algebra, and every nonzero element is invertible.

Set

$$P_0 = 0, \quad P = 1, \quad C = 0.$$

Use the normalized coprime factorizations

$$P_0 = \frac{0}{1}, \quad P = \frac{1/\sqrt{2}}{1/\sqrt{2}}, \quad C = \frac{0}{1}.$$

Thus

$$G_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \tilde{G}_0 = \begin{bmatrix} -1 & 0 \end{bmatrix}, \quad G = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}.$$

The determinant condition in the definition of d_{ν} holds because

$$G^* G_0 = \frac{1}{\sqrt{2}} \neq 0$$

and the index is trivial. Consequently

$$d_\nu(P, P_0) = \|\tilde{G}_0 G\|_\infty = \left| -\frac{1}{\sqrt{2}} \right| = \frac{1}{\sqrt{2}} > \frac{1}{2}.$$

On the other hand, the source's closed-loop matrix is

$$H(Q, C) = \begin{bmatrix} Q \\ 1 \end{bmatrix} (1 - CQ)^{-1} \begin{bmatrix} -C & 1 \end{bmatrix}.$$

For the nominal pair,

$$H(P_0, C) = H(0, 0) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad \|H(0, 0)\| = 1,$$

so C stabilizes P_0 and $\mu_{P_0, C} = 1 > \frac{1}{2}$. The same controller also stabilizes $P = 1$, since

$$H(P, C) = H(1, 0) = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \in R^{2 \times 2}.$$

Hence both hypotheses on the left side of (1) hold with $m = \frac{1}{2}$, while its conclusion fails. This proves the claim. $\square$

Interpretation boundary

The example uses constant transfer functions, so it also belongs to the rational class that the source says already satisfies (P'). Therefore the displayed sentence cannot simultaneously have its literal implication meaning and the asserted rational-case provenance. Standard robust stability says that a controller with margin exceeding a radius stabilizes *every plant in the corresponding ν -ball*; it does not say that a controller can stabilize no plant outside that ball. Indeed, the zero controller stabilizes every stable scalar plant, including many plants far from the zero plant.

Thus the open question as printed has a complete negative answer. If the authors intended a worst-case statement about an entire uncertainty ball, or a minimality statement among metrics, that is a different quantified problem and is not refuted by this packet.

References

- [1] J. A. Ball and A. J. Sasane, *Extension of the ν -metric*, Complex Analysis and Operator Theory 6 (2012), 65–89, arXiv:1003.0108, <https://arxiv.org/abs/1003.0108>.